LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	20
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Ice and rain protection systems, methods of protection, pneumatic de-icing system, electrical de-icing and anti-icing system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Ice and rain protection systems, methods of protection, pneumatic de-icing system, electrical de-icing and anti-icing system.
5. Specific Objectives	13.1.3.1 Knowledge on circuit protection devices, generators, actuators & engine starting system.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of IB schematics for pneumatic and electrical de-icing loops; verify circuit breaker locations and actuator power sources; instructor supervises schematic tracing and component identification.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary hazards of ice accumulation on aircraft surfaces?
(b) How does a pneumatic de-icing system utilize engine bleed air?
(c) What is the fundamental difference between de-icing and anti-icing methods?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Ice formation types and critical surfaces • Methods of protection overview • Pneumatic de-icing system operation • Engine bleed air supply and distribution	L & Ds	PPT, IB	15 min
(b)	• Electrical de-icing system components • Heating element types and placement • Circuit protection devices and generators • Actuator operation in electrical systems	L & Ds	PPT, PDF	20 min
(c)	• Anti-icing system integration • System control logic and sequencing • Maintenance checks for de-icing systems • Troubleshooting common electrical faults	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of circuit protection devices in the electrical de-icing loop.
(b) Describe how the engine starting system relates to pneumatic de-icing availability.
(c) Differentiate between the operational requirements of de-icing versus anti-icing systems.

12. Home Assignment and Follow Up	Prepare a brief note comparing the advantages and limitations of pneumatic versus electrical de-icing systems for future application in LP 21 regarding engine air intake protection.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Engine air intake and propeller systems, windshield anti-icing systems, ice detection.	Refer to the master syllabus reference list	LP 21

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min